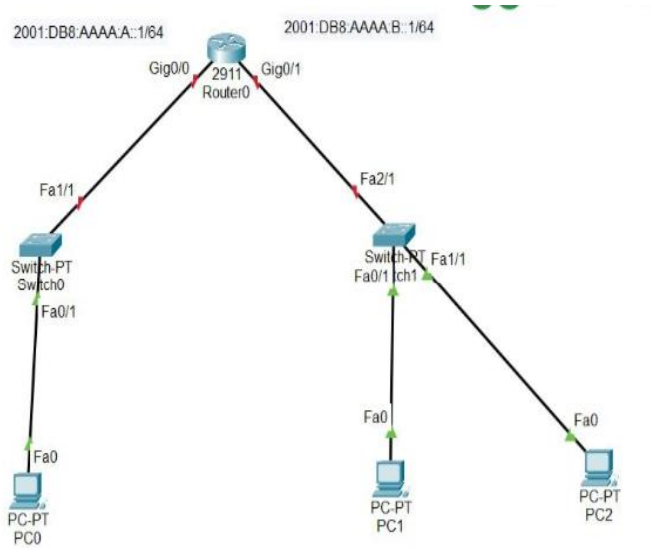
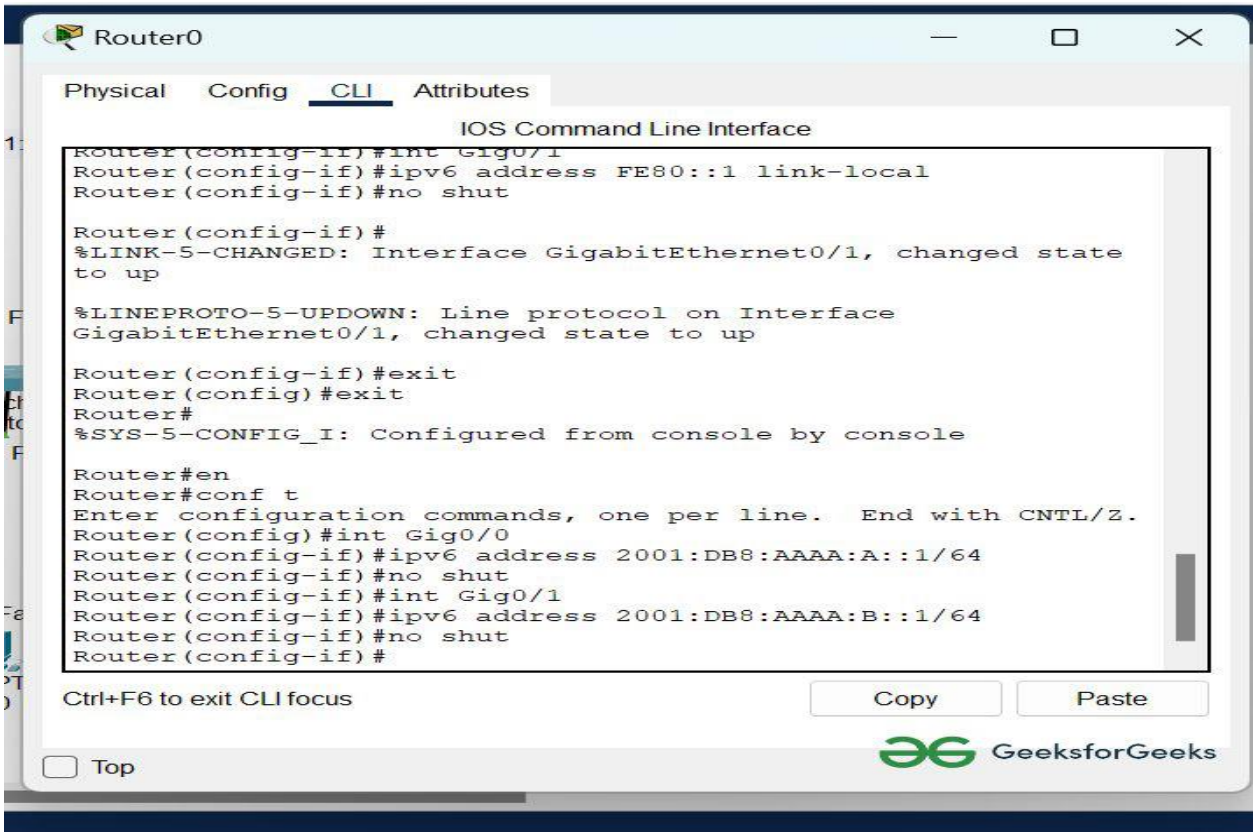
Practical 6: IPv6 and ICMP Tools

a. **AIM:** Configure IPv6 addresses in Packet Tracer.





PC1

Physical Config Desktop Programming Attributes

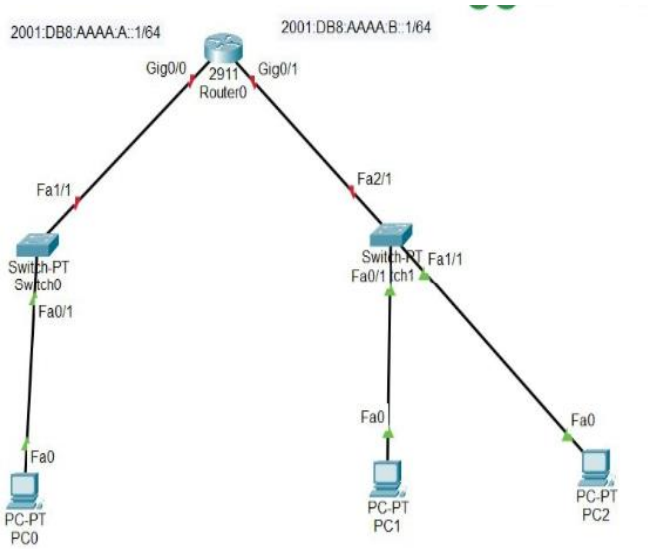
Command Prompt

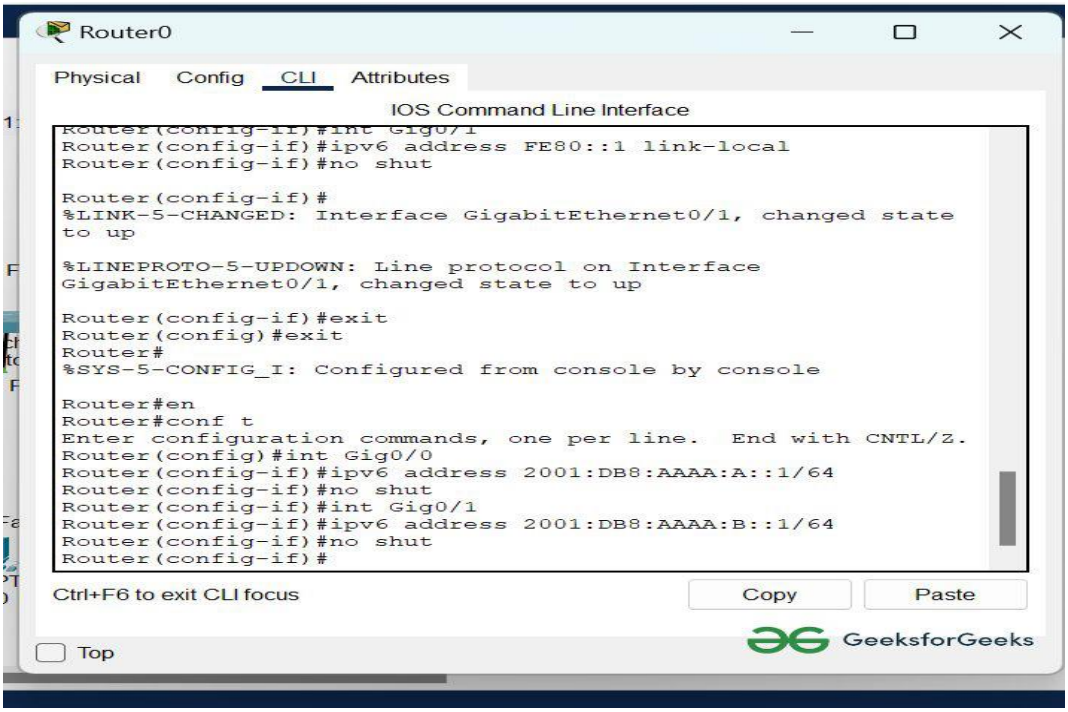
```
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 2001:DB8:AAAA:A:20D:BDFF:FE1A:D121:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56

Pinging 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56 with 32 bytes of
data:

Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time=10ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time<1ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time=3ms TTL=127
Reply from 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56: bytes=32
time<1ms TTL=127

Ping statistics for 2001:DB8:AAAA:A:207:ECFF:FEA3:EB56:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>
```

Top

GeeksforGeeks

b. Use ping6 and traceroute6 to test connectivity.





c. **AIM:** Analyze ICMP error and echo messages with Wireshark

```
Internet Control Message Protocol v6
    Type: Neighbor Solicitation (135)
    Code: 0
    Checksum: 0x28de [correct]
    [Checksum Status: Good]
    Reserved: 00000000
    Target Address: fe80::1
    ICMPv6 Option (Source link-layer address : c8:53:09:63:4e:06)
        Type: Source link-layer address (1)
        Length: 1 (8 bytes)
        Link-layer address: LCFCElectron_63:4e:06 (c8:53:09:63:4e:06)
```

d. **AIM:** Compare IPv4 and IPv6 header

```
Internet Protocol Version 4, Src: 192.168.1.15, Dst: 23.212.254.56
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        0000 00.. = Differentiated Services Codepoint: Default (0)
        .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    Total Length: 1424
    Identification: 0x4c02 (19458)
    010. .... = Flags: 0x2, Don't fragment
        0... .... = Reserved bit: Not set
        .1.. .... = Don't fragment: Set
        ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: TCP (6)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.15
    Destination Address: 23.212.254.56
    [Stream index: 26]
```

```
Internet Protocol Version 6, Src: fe80::7c2b:9ebf:65dd:21be, Dst: ff02::16
    0110 .... = Version: 6
    .... 0000 0000 .... .... .... .... .... = Traffic Class: 0x00 (DSCP: CS0, ECN: Not-ECT)
        .... 0000 00.. .... .... .... .... .... = Differentiated Services Codepoint: Default (0)
        .... .... ..00 .... .... .... .... .... = Explicit Congestion Notification: Not ECN-Capable Transport (0)
    .... 0000 0000 0000 0000 0000 = Flow Label: 0x00000
    Payload Length: 36
    Next Header: IPv6 Hop-by-Hop Option (0)
    Hop Limit: 1
    Source Address: fe80::7c2b:9ebf:65dd:21be
        [Address Space: Link-Local Unicast]
        [Special-Purpose Allocation: Link-Local Unicast]
    Destination Address: ff02::16
        [Address Space: Multicast]
        [.... .... 0000 .... = Multicast Flags: 0x0]
            .... .... 0... .... = Reserved: 0
            .... .... .0.. .... = Rendezvous Point (RP): False
            .... .... ..0. .... = Network Prefix: False
            .... .... ...0 .... = Transient: False
        [.... .... .... 0010 = Multicast Scope: Link-Local scope (0x2)]
    [Stream index: 18]
    IPv6 Hop-by-Hop Option
        Next Header: ICMPv6 (58)
        Length: 0
        [Length: 8 bytes]
        Router Alert
            Type: Router Alert (0x05)
            Length: 2
            Router Alert: MLD (0)
        PadN
            Type: PadN (0x01)
                00.. .... = Action: Skip and continue (0)
                ..0. .... = May Change: No
                ...0 0001 = Low-Order Bits: 0x01
            Length: 0
            PadN: <none>
```

Practical 07

Practical 7: Transport Layer Protocol Testing

a. **AIM:** Use Wireshark to analyze a TCP three-way handshake.

```
Transmission Control Protocol, Src Port: 51302, Dst Port: 7680, Seq: 0, Len: 0
    Source Port: 51302
    Destination Port: 7680
    [Stream index: 3]
    [Stream Packet Number: 4]
  > [Conversation completeness: Incomplete, SYN_SENT (1)]
    [TCP Segment Len: 0]
    Sequence Number: 0    (relative sequence number)
    Sequence Number (raw): 526380124
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 0
    Acknowledgment number (raw): 0
    1000 .... = Header Length: 32 bytes (8)
  > Flags: 0x002 (SYN)
    Window: 65535
    [Calculated window size: 65535]
    Checksum: 0xf8ea [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  > Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No-Operation (NOP), SACK permit
  > [Timestamps]
  > [SEQ/ACK analysis]
    [Client Contiguous Streams: 0]
    [Server Contiguous Streams: 1]
```

b. **AIM:** Numerical on UDP and TCP header

UDP dump Structure

5M Q. XX XX XX XX XX XX XX XX

Source port | Destination port | length | checksum

given a dump of UDP header in hexa format

06 32 00 0D 00 1C E2 17 find the following

a) Source code number
b) Destination — " —
c) Length of user datagram
d) Length of data.
e) UDP header diagram

Solution:-

UDP dump structure -

XX XX XX XX XX XX XX XX

Source port | Destination port | length | checksum

a) Source code -

It is port that identify number from sender side. It helps reciever to understand which application is sending the data.

It is a 2 ~~bit~~ byte (16 bits) long field.

for the given dump the source code

$2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$

16 8 4 2 1

0 1 1 0

is equal to 06 32

$(06\ 32)_{16} =$

1024
512
32
16
2
1586

$(06\ 32)_{16} = \cancel{(812)_{10}}\ (1586)_{10}$

0000 0110 0011 0010

0110
0011
0010

B = 1-0
D = 1-9

10 - A
11 - B
12 - C
13 - D

(2) Destination port :-

$(00\ 0D)_{10} = 13$

(3) Length :-

28

Size Header = 8 bytes
Total length = 28 bytes
length of data = Total length − header
= 28 − 8
= 20 bytes

Data travers = Client → Servers

Source (16 bits)	Destination port (16 bits)
length 16 bits	Checksum 16 bits
8 byts	Data

Q. 053200217 000000001 000000000 500207FF
000000000

TCP Header diagram :-

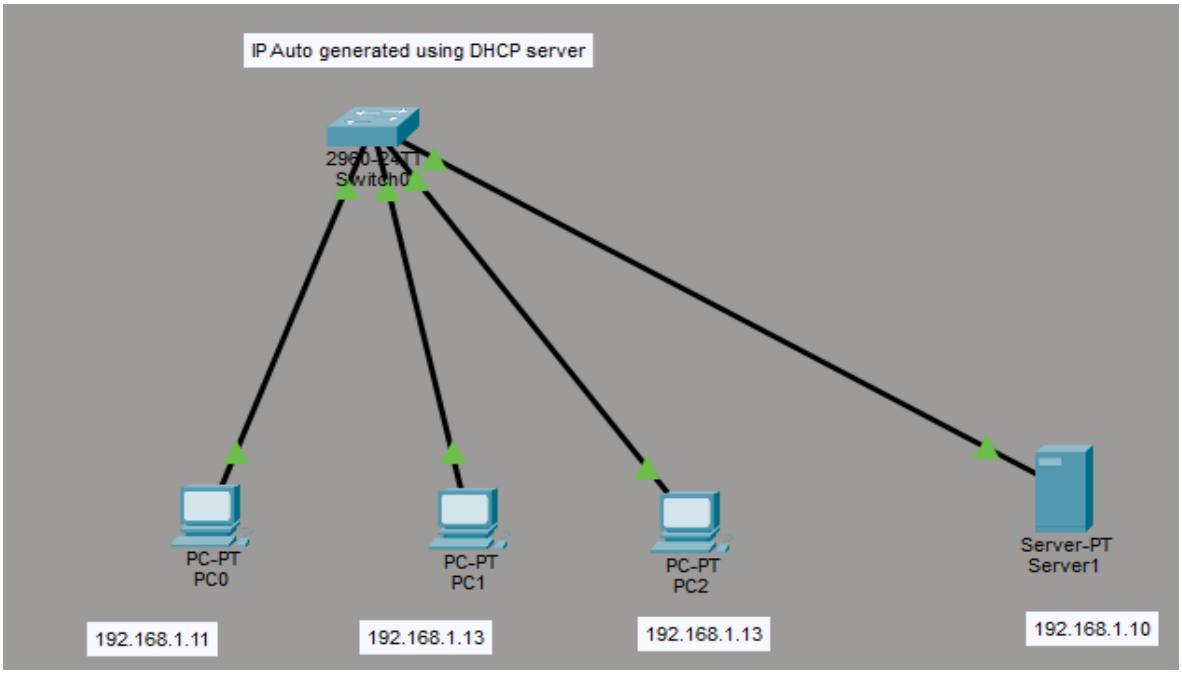
Source 16					Destination 16			
Sequence Number 32								
Acknowledgment Number 32								
HLen 4 bits	Round 6	URG	ACK	PSH	RST	SYN	FIN	windowsize 16
Checksum 16								Urgent point 16
optional Padding (40 bytes)								

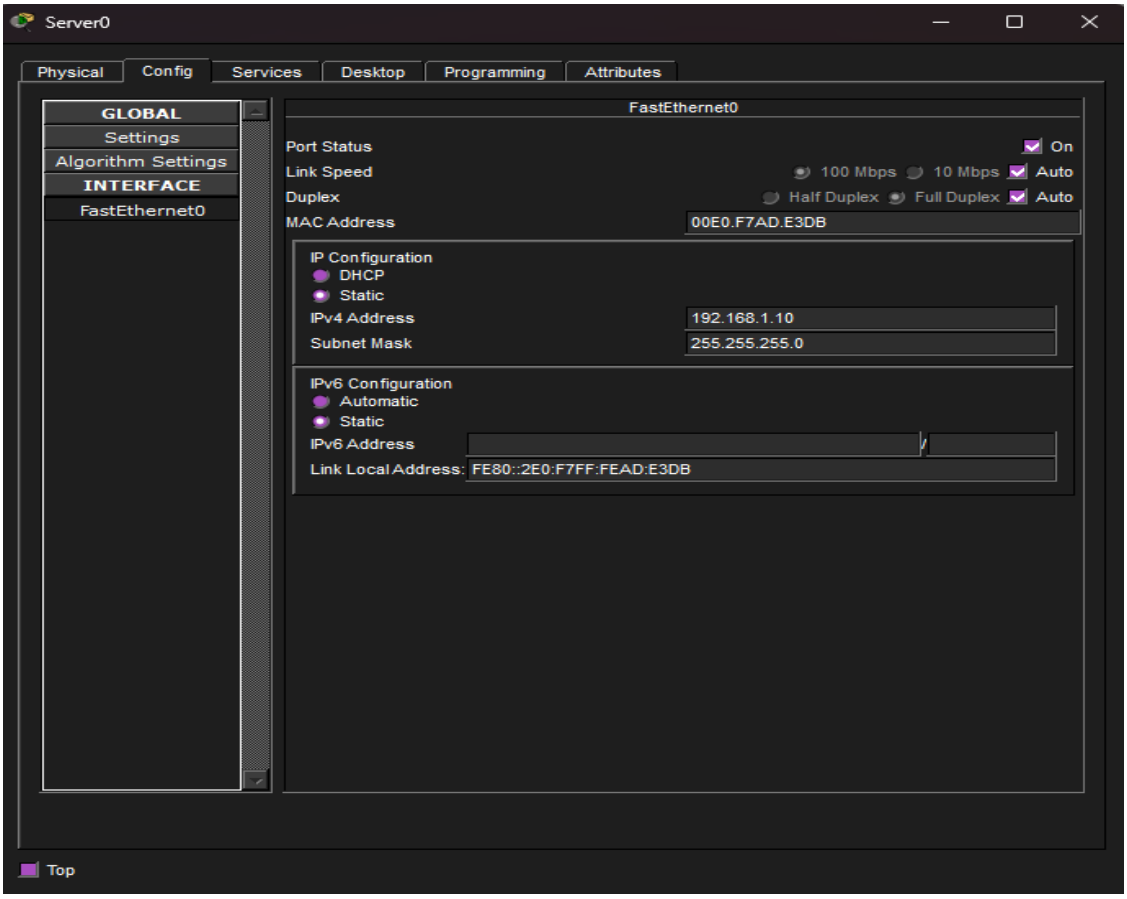
TCP dump header structure.

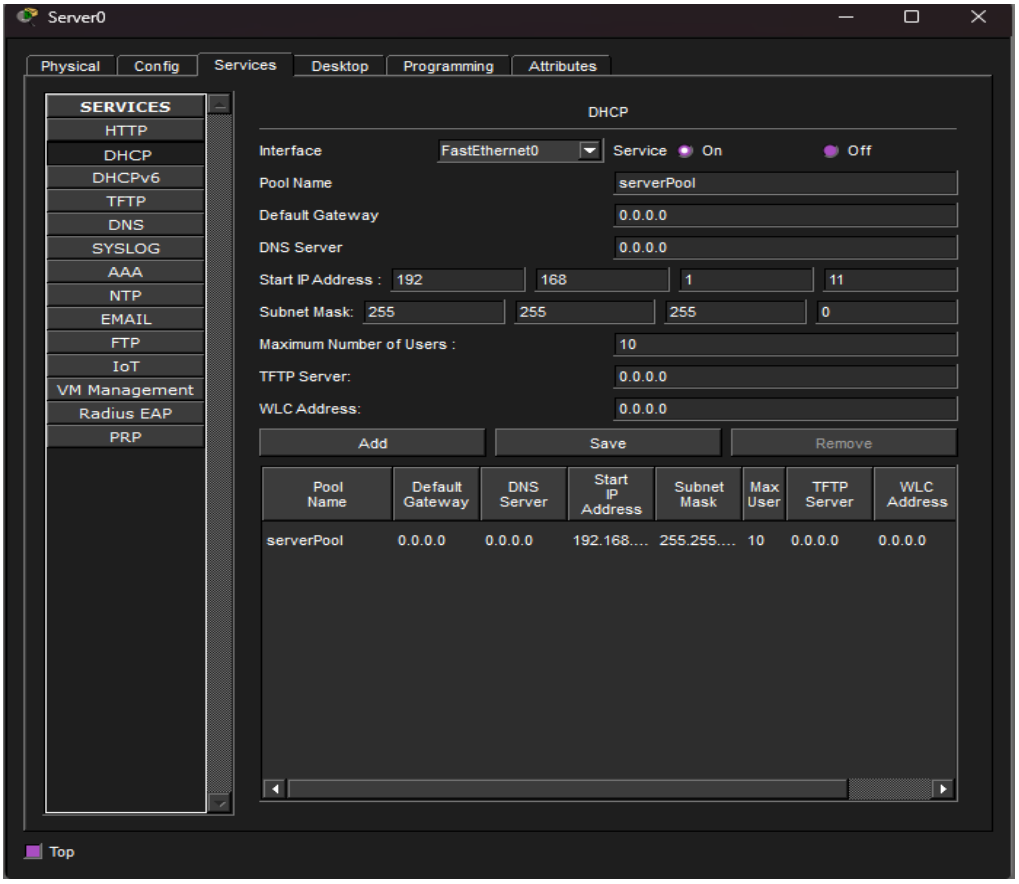
Practical 08

Practical 8: DHCP and IP Allocation Technique

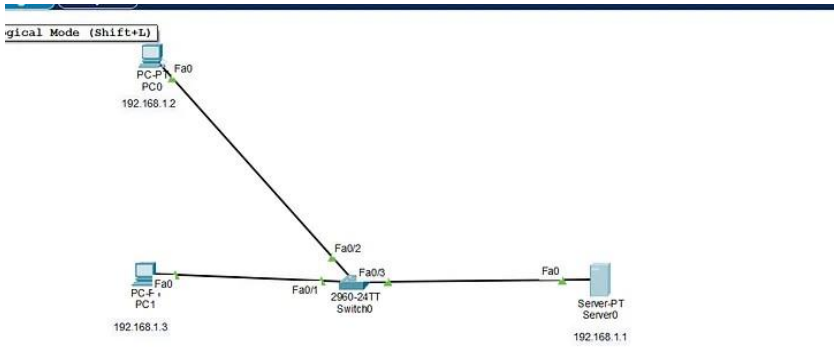
a. **AIM: Configure a DHCP server in Cisco Packet Tracer.s**

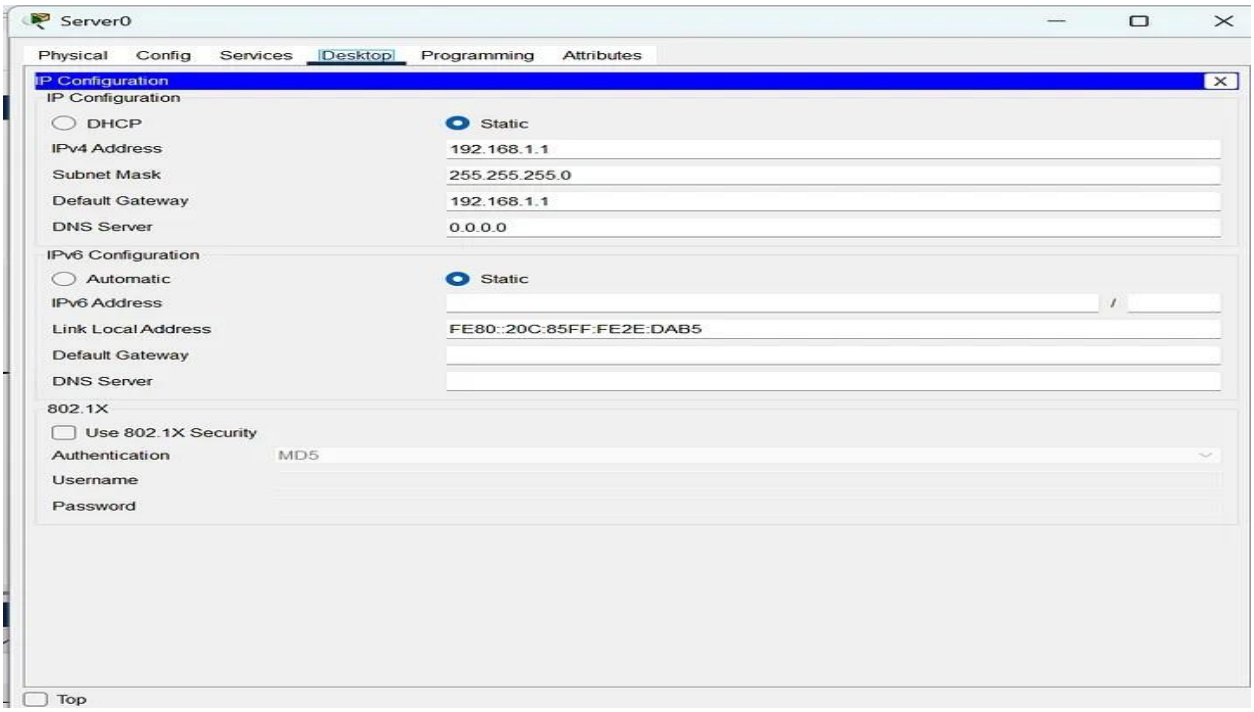



Server0
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
FastEthernet0
Port Status
On
Link Speed
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F7AD.E3DB
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address:
FE80::2E0:F7FF:FEAD:E3DB
Top


Server0
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DHCP
Interface
FastEthernet0
Service
On
Off
Pool Name
serverPool
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
Start IP Address :
192
168
1
11
Subnet Mask:
255
255
255
0
Maximum Number of Users :
10
TFTP Server:
0.0.0.0
WLC Address:
0.0.0.0
Add
Save
Remove
Pool Name
Default Gateway
DNS Server
Start IP Address
Subnet Mask
Max User
TFTP Server
WLC Address
serverPool
0.0.0.0
0.0.0.0
192.168....
255.255....
10
0.0.0.0
0.0.0.0
Top

b. **AIM:** Assign IP addresses dynamically to multiple clients and verify using ipconfig.





PC0

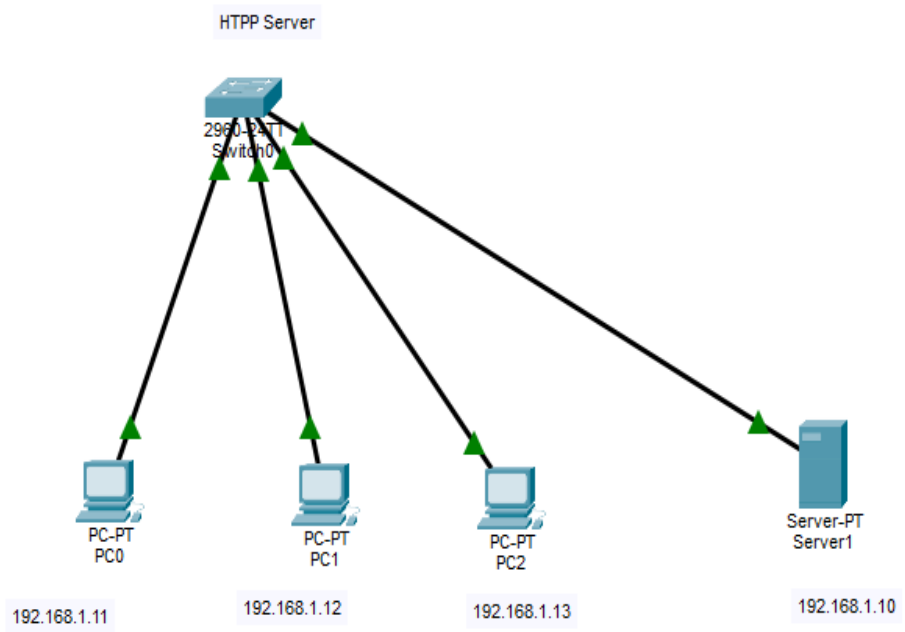
Physical Config Desktop Programming Attributes

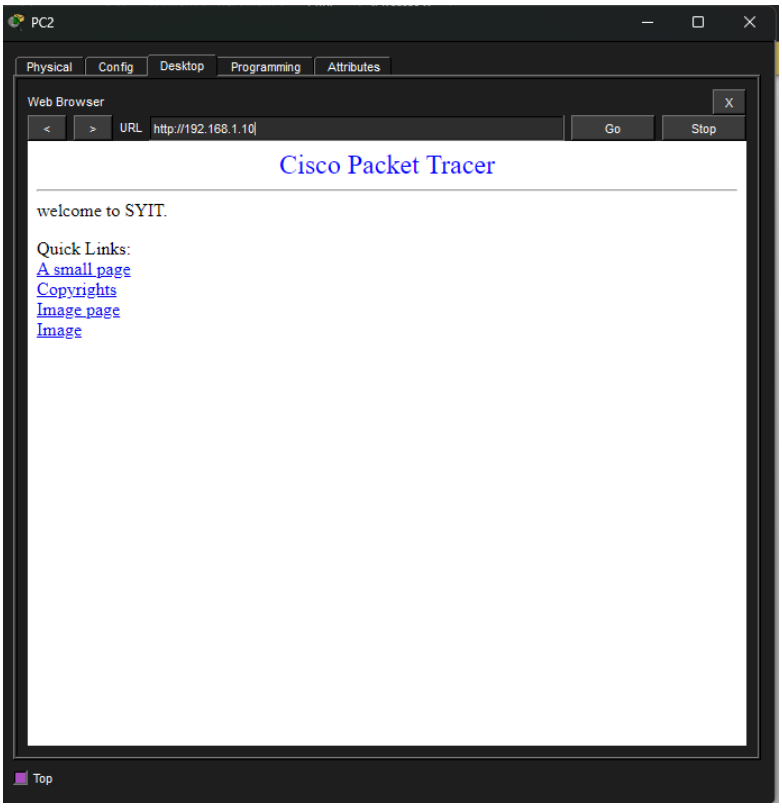
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

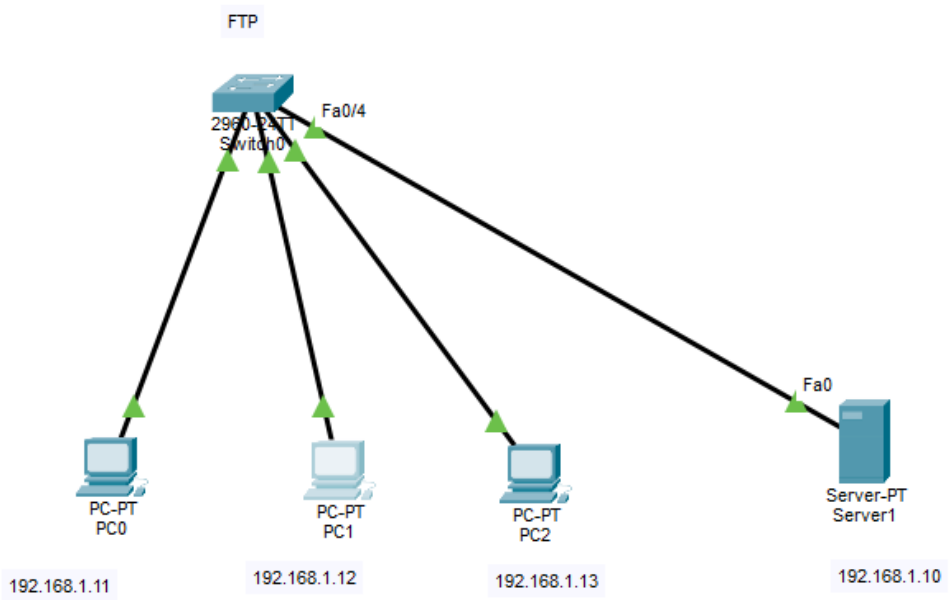
Practical 9: Application Layer Protocols

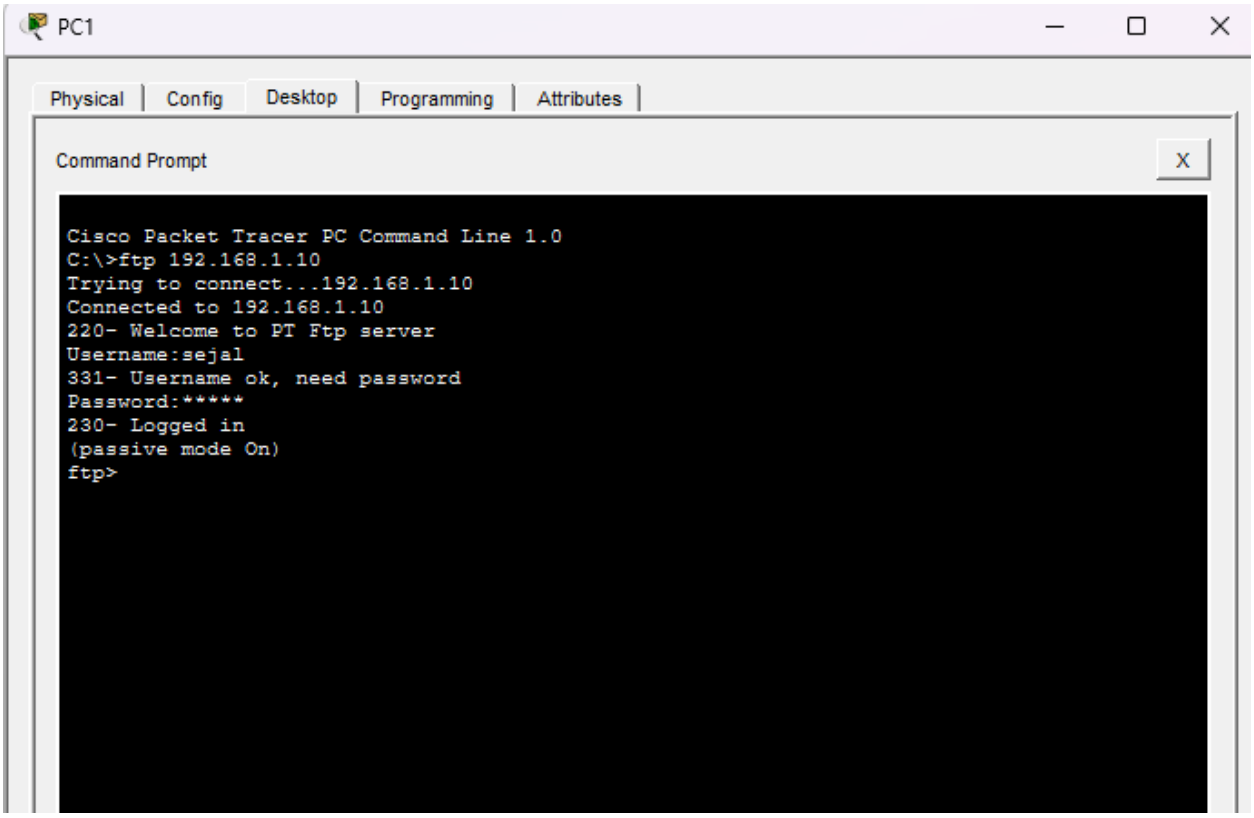
a. AIM: Set up DNS SeS Server





b. AIM: . Set up an FTP server and download/upload files.





PC2

Physical | Config | Desktop | Programming | Attributes

Command Prompt

```
ftp>quit

221- Service closing control connection.
C:\>ftp 192.168.1.10
Trying to connect...192.168.1.10
Connected to 192.168.1.10
220- Welcome to PT Ftp server
Username:sejal
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 192.168.1.10:
0    : asa842-k8.bin                                    5571584
1    : asa923-k8.bin                                    30468096
2    : c1841-advipservicesk9-mz.124-15.T1.bin           33591768
3    : c1841-ipbase-mz.123-14.T7.bin                    13832032
4    : c1841-ipbasek9-mz.124-12.bin                     16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin           33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin           33591768
7    : c2600-i-mz.122-28.bin                            5571584
8    : c2600-ipbasek9-mz.124-8.bin                      13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin         50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin          33591768
11   : c2800nm-ipbase-mz.123-14.T7.bin                  5571584
12   : c2800nm-ipbasek9-mz.124-8.bin                    15522644
13   : c2900-universalk9-mz.SPA.155-3.M4a.bin           33591768
14   : c2950-i6q412-mz.121-22.EA4.bin                   3058048
15   : c2950-i6q412-mz.121-22.EA8.bin                   3117390
16   : c2960-lanbase-mz.122-25.FX.bin                   4414921
17   : c2960-lanbase-mz.122-25.SEE1.bin                 4670455
18   : c2960-lanbasek9-mz.150-2.SE4.bin                 4670455
19   : c3560-advipservicesk9-mz.122-37.SE1.bin          8662192
20   : c3560-advipservicesk9-mz.122-46.SE.bin           10713279
21   : c800-universalk9-mz.SPA.152-4.M4.bin             33591768
22   : c800-universalk9-mz.SPA.154-3.M6a.bin            83029236
23   : cat3k_caa-universalk9.16.03.02.SPA.bin           505532849
24   : cgr1000-universalk9-mz.SPA.154-2.CG              159487552
25   : cgr1000-universalk9-mz.SPA.156-3.CG              184530138
26   : ie9k_iosxe.17.09.04.SPA.bin                      596133776
```

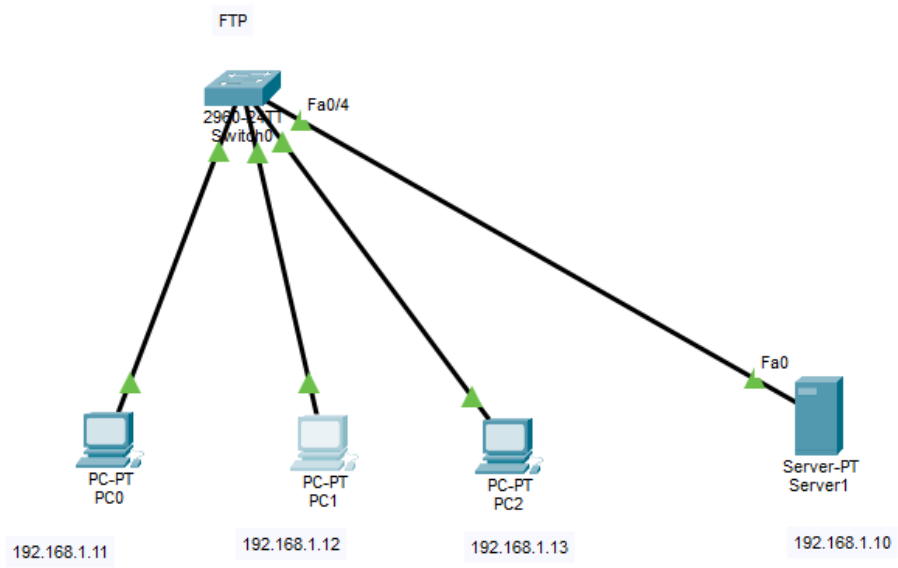
Top

c. **AIM:** Configure a basic SMTP and POP3 email exchange (simulation).

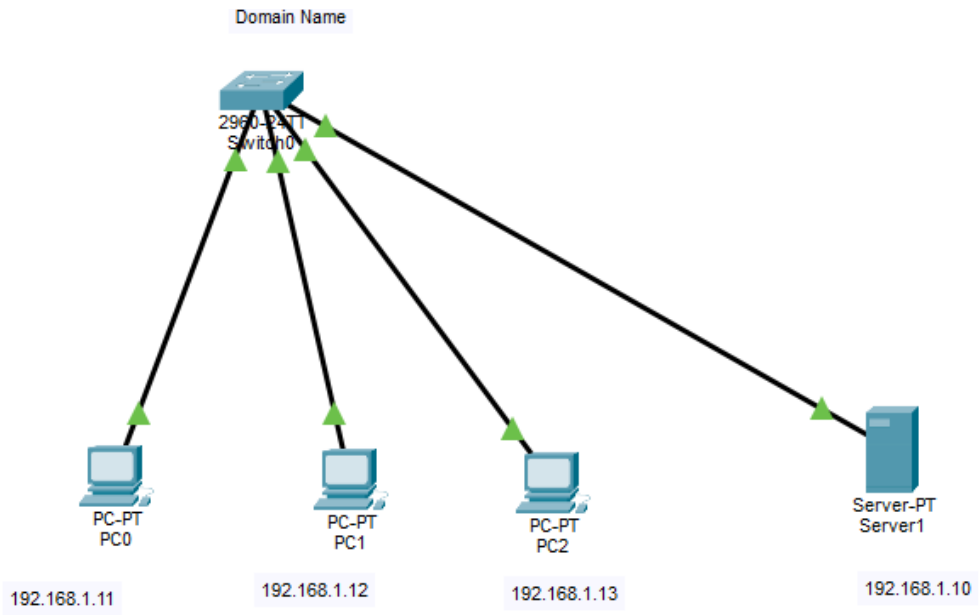
IOS Command Line Interface

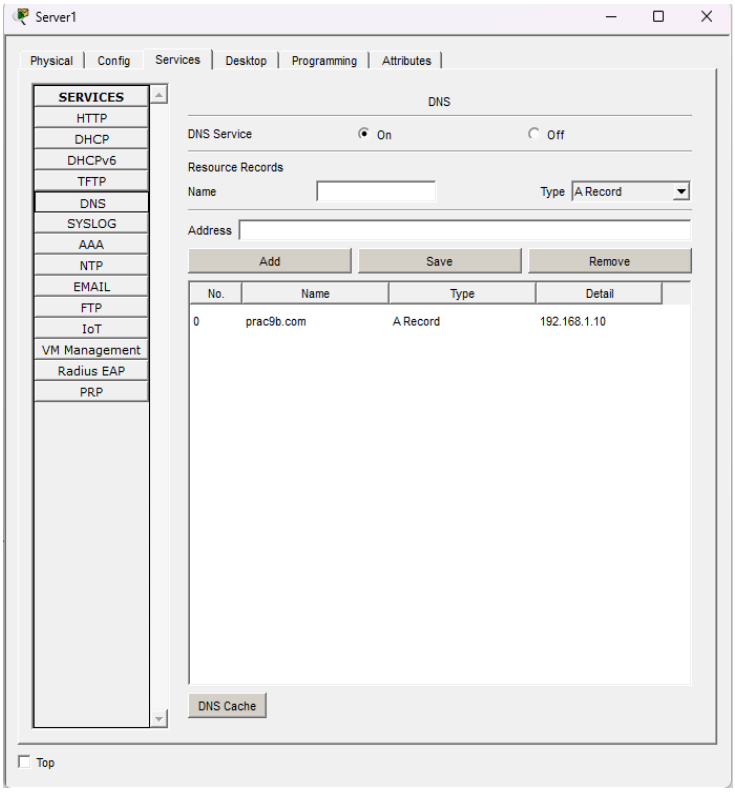
```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed sta
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#int
Router(config)#interface gi
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip add 192.168.1.100 255.255.255.0
Router(config-if)#no shut
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed sta
to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface
GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

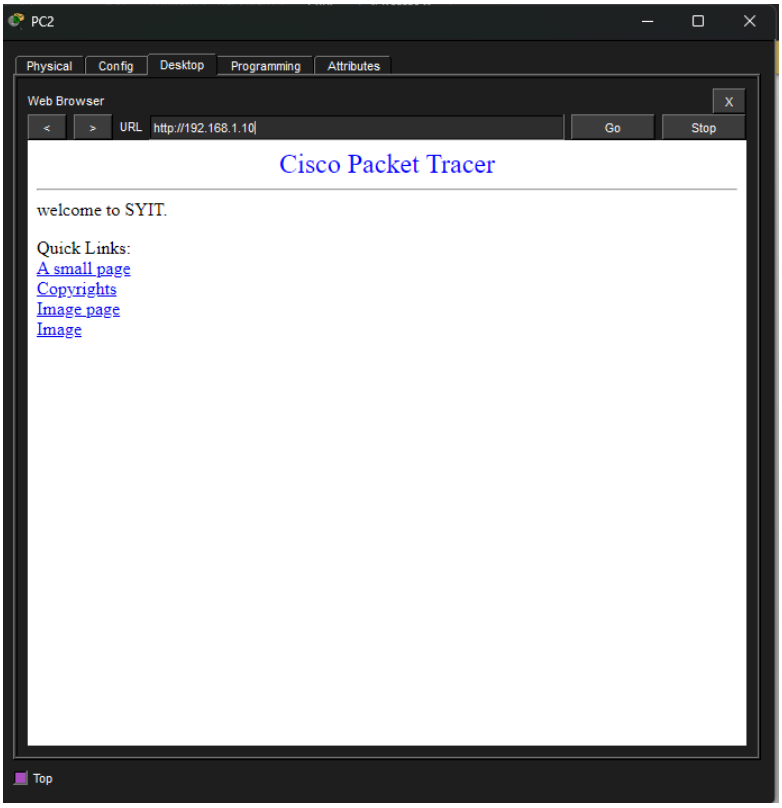
Copy



d. **AIM:** Access a website and inspect HTTP headers using browser tools.

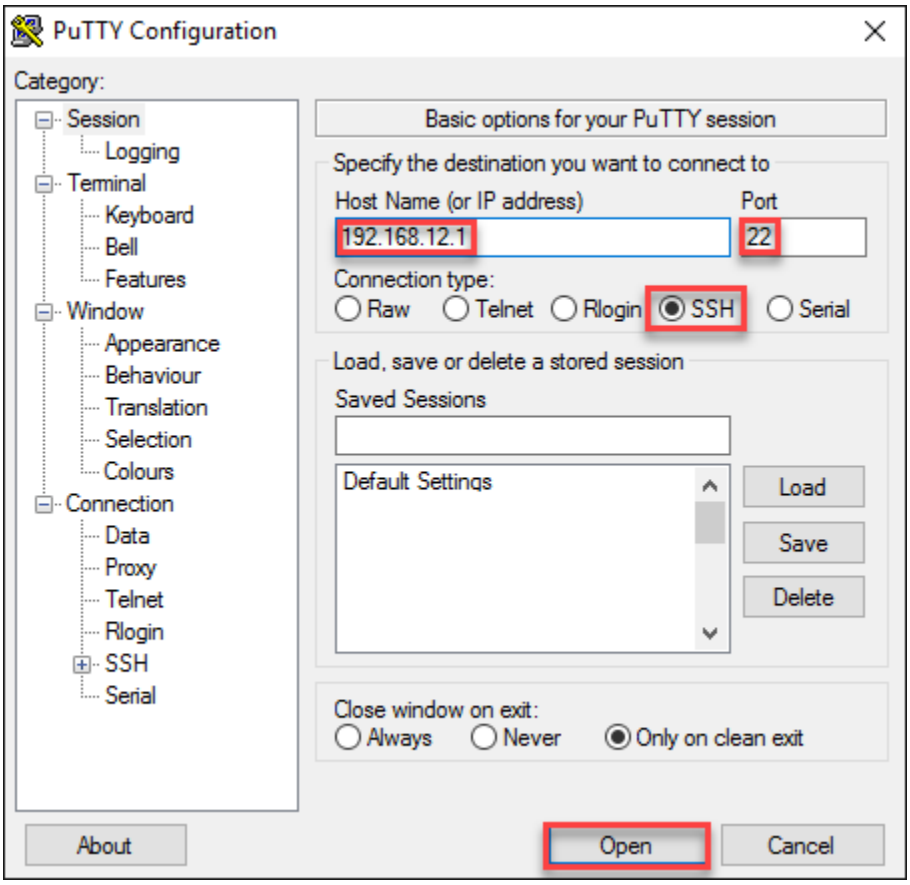





PC2
Physical
Config
Desktop
Programming
Attributes
Web Browser
X
<
>
URL
http://192.168.1.10
Go
Stop
Cisco Packet Tracer
welcome to SYIT.
Quick Links:
A small page
Copyrights
Image page
Image
Top

Practical 10: Remote Access and Secure Connections

a. **AIM:** Connect to a remote server using TELNET (simulation).

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>telnet
Cisco Packet Tracer PC Telnet

Usage: telnet target [port]

C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open


User Access Verification

Username: cisco
Password:
% Password:  timeout expired!
% Login invalid

[Connection to 192.168.1.1 closed by foreign host]
C:\>
```

b. Establish an SSH connection with authentication.